



Assignment 1

Problem Statement:

LAN setup & packet analysis with Network commands Testing and File Transmission:
To design and Configure a LAN, analyze network traffic using Wireshark packet analyzer tool, test connectivity with standard network commands & demonstrate file transmission between system.

Objectives:

1. To understand basics of Computer Networking.
2. To setup LAN
3. To understand the working of common network protocol.
4. To analyze network packets using Wireshark
5. To demonstrate successful file transfer between systems connected to LAN.

Requirements:

Hardware :

1. Two or more computers.
2. Ethernet cables
3. Network switch

Software :

1. Wireshark packet Analyzer
2. Windows/Linux OS.
3. File sharing utility.
(FTP/SCP/SMB)

Theory:

4.1 : ~~Network~~ Computer Network.

A computer network is a collection of interconnected devices that communicate with each other to share data, resources & services.



4.2 Types of Network:-

A network enables communication between multiple computing devices using communication links.

Depending on geographical coverage, networks are classified as:

1. PAN (Personal Area Network) - covers a few meters used for Bluetooth.
2. LAN (Local Area Network) - covers building or campus.
3. MAN (Metropolitan Area Network) - covers an entire city.
4. WAN (Wide Area Network) - covers countries & continents.

4.3. Network Protocols.

Rules governing communication betⁿ devices.

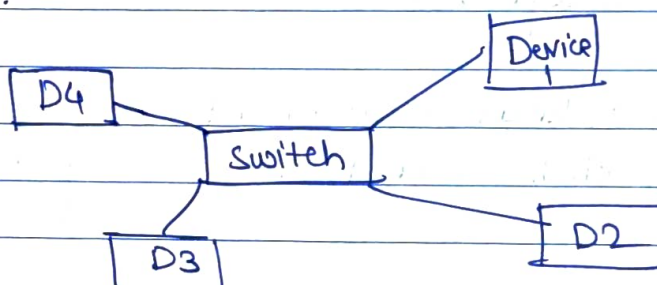
1. TCP: • Reliable connection oriented protocol
• Ensures ordered & error free delivery.
2. UDP: • Connectionless Protocol
• Faster than TCP
• Used in DHCP, DNS, VoIP & streaming
3. ARP (Address Resolution Protocol):
• Maps an IP Address to a MAC address.
• Enables communication within a LAN.
4. DHCP (Dynamic Host Configuration Protocol):
• Automatically assigns IP address to devices joining a network



5. mDNS (Multicast DNS): Resolves local device names without requiring a DNS server
6. SSL/TLS :
 - Provides encryption & secure communication
 - Operates over TCP.

4.4. Network Topologies:

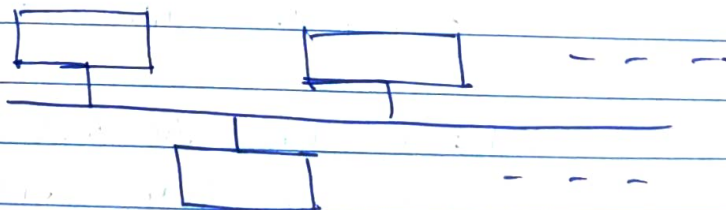
1) Star Topology:



Failure of one cable affects only one device.
Switch failure disconnects the entire network.

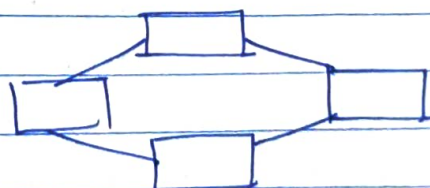
2) Bus Topology

- Single backbone cable
- Backbone failure of entire network.



3) Ring Topology :-

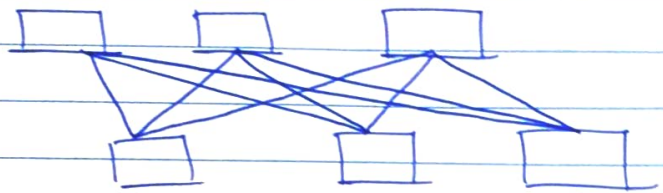
- Devices connected in circular fashion.
- Data travels in one or both directions.





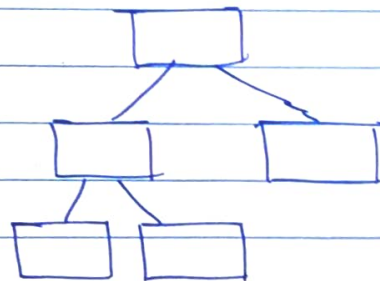
- A broken link can disrupt communication.

4) Mesh Topology:



- Every device connects to every other device
- Highly reliable
- Expensive due to many links.

5) Tree Topology



- Hierarchical arrangement of switches & routers.

4.5) Network Devices:-

- ① Hub :
 - Broadcasts incoming data to every connected device
 - Generates unnecessary traffic and collisions.
- ② Switch :
 - Learns MAC Address.
 - Sends frames only to intended devices.
- ③ Router :
 - Connects different networks.
 - Uses routing tables to forward packets towards their destination.
- ④ Repeater :
 - Regenerates weakened electrical signals
 - Extends network communication distance.



4.6. Ethernet : Ethernet is most widely used LAN technology. It defines standards for transmitting data over wired networks using ethernet cables.

Ethernet uses MAC addresses for communication within a local network & supports high speed, reliable data transmission through switches.

* Wireshark : A packet analyzer used to capture & inspect network traffic.

Common Protocols:

- ARP
- TCP
- UDP
- DHCP
- mDNS
- TLS/SSL

Useful Display filters:
ip.addr == 10.10.15.22

ip route

Displays the routing table & default gateway.

ping <IP Address>

- Checks connectivity with another host
- Packet capture confirms successful packet exchange between communicating systems.

* Conclusion :-

- The experiment successfully demonstrated the design & configuration of a LAN.

Packet Analysis using Wireshark provided insight into the operations of protocols such as ARP, TCP, UDP, etc.